

2010-2011 工科数学分析第二学期期末试题 (A 卷) 解答 (2011. 6)

一. 1. -30

2. $1+e$

3. $-\frac{1}{2}$

4. $\int_L \frac{x^2 + 3x^2 y}{\sqrt{1+9x^4}} dl$

5. $u \frac{\partial z}{\partial u} = z$

二. $I = \int_0^1 \frac{e^x}{x} dx \int_{x^2}^x dy \dots\dots\dots(3 \text{ 分})$

$= \int_0^1 (e^x - xe^x) dx \dots\dots\dots(6 \text{ 分})$

$= e - 2 \dots\dots\dots(9 \text{ 分})$

三. $f'_x = 2xy$, $f'_y = x^2 + y - 1 \dots\dots\dots(2 \text{ 分})$

令 $f'_x = 0$, $f'_y = 0$, 得 $x = 0, y = 1$ 或 $y = 0, x = \pm 1$

得三点 $P_1(0,1)$, $P_2(1,0)$, $P_3(-1,0) \dots\dots\dots(4 \text{ 分})$

$f''_{x^2} = 2y$, $f''_{xy} = 2x$, $f''_{y^2} = 1 \dots\dots\dots(5 \text{ 分})$

在点 P_1 , $A = 2, B = 0, C = 1$, $AC - B^2 = 2 > 0$, $A = 2 > 0$

$P_1(0,1)$ 是极小点, 极小值 $f(0,1) = -\frac{1}{2} \dots\dots\dots(7 \text{ 分})$

在点 P_2 , $A = 0, B = 2, C = 1$, $AC - B^2 = -4 < 0$,

在点 P_3 , $A = 0, B = -2, C = 1$, $AC - B^2 = -4 < 0$,

P_2, P_3 都不是极值点 $\dots\dots\dots(9 \text{ 分})$

四. $3z^2 \frac{\partial z}{\partial x} - 2z - 2x \frac{\partial z}{\partial x} = 0$ (2 分)

$$\frac{\partial z}{\partial x} = \frac{2z}{3z^2 - 2x}$$
(3 分)
$$3z^2 \frac{\partial z}{\partial y} - 2x \frac{\partial z}{\partial x} + 1 = 0$$
(5 分)
$$\frac{\partial z}{\partial y} = \frac{-1}{3z^2 - 2x},$$
(6 分)
$$\frac{\partial^2 z}{\partial x \partial y} = \frac{2 \frac{\partial z}{\partial y} (3z^2 - 2x) - 2z(6z) \frac{\partial z}{\partial y}}{(3z^2 - 2x)^2}$$
(7 分)
$$= \frac{6z^2 + 4x}{(3z^2 - 2x)^3}$$
(9 分)

五. 设切点 $M(x_0, y_0, z_0)$, $\frac{\partial z}{\partial x} = y$, $\frac{\partial z}{\partial y} = x$, 法向量 $\vec{n} = \{y_0, x_0, -1\}$ (3 分)

由题设, $\vec{n} // \{1, 3, 1\}$, $\frac{y_0}{1} = \frac{x_0}{3} = \frac{-1}{1}$ (5 分)

得 $x_0 = -3$, $y_0 = -1$, $z_0 = x_0 y_0 = 3$, 所求点为 $M(-3, -1, 3)$ (7 分)

切平面为 $(x+3) + 3(y+1) + (z-3) = 0$

即 $x + 3y + z + 3 = 0$ (9 分)

六. $\frac{\partial Y}{\partial x} = \frac{\partial X}{\partial y} = x^{y-1} + yx^{y-1} \ln x$, 故是全微分方程 (2 分)

$$u(x, y) = \int_{(1,0)}^{(x,y)} yx^{y-1} dx + x^y \ln x dy$$
 (4 分)

$$= \int_1^x 0 dx + \int_0^y x^y \ln x dy$$
 (6 分)

$$= x^y - 1$$
 (7 分)

通解为 $x^y = C$ (8 分)

七. $\lim_{n \rightarrow \infty} \left| \frac{a_{n+1}}{a_n} \right| = 1, R = 1$ (1 分)

$x = \pm 1$ 时, 级数为 $\sum_{n=1}^{\infty} n(n+1)(\pm 1)^{n-1}$, 发散, 收敛域为 $(-1, 1)$ (2 分)

设 $S(x) = \sum_{n=1}^{\infty} n(n+1)x^{n-1}$

$$\int_0^x S(x) dx = \sum_{n=1}^{\infty} (n+1)x^n \quad \dots\dots\dots (4 \text{ 分})$$

$$\int_0^x \left(\int_0^x S(x) dx \right) dx = \sum_{n=1}^{\infty} x^{n+1} \quad \dots\dots\dots (6 \text{ 分})$$

$$= \frac{x^2}{1-x} = \frac{1}{1-x} - 1 - x \quad \dots\dots\dots (8 \text{ 分})$$

$$\int_0^x S(x) dx = \left(\frac{1}{1-x} - 1 - x \right)' = \frac{1}{(1-x)^2} - 1 \quad \dots\dots\dots (9 \text{ 分})$$

$$S(x) = \left(\frac{1}{(1-x)^2} - 1 \right)' = \frac{2}{(1-x)^3} \quad \dots\dots\dots (10 \text{ 分})$$

八. $m = \iiint_V \frac{1}{x^2 + y^2 + z^2} dV$ (1 分)

$$= \int_0^{2\pi} d\theta \int_0^{\frac{\pi}{4}} d\varphi \int_0^{2\cos\varphi} \frac{1}{r^2} r^2 \sin\varphi dr \quad \dots\dots\dots (3 \text{ 分})$$

$$= 4\pi \int_0^{\frac{\pi}{4}} \sin\varphi \cos\varphi d\varphi = \pi \quad \dots\dots\dots (4 \text{ 分})$$

$$\bar{x} = \bar{y} = 0, \quad \dots\dots\dots (6 \text{ 分})$$

$$\bar{z} = \frac{1}{m} \iiint_V \frac{z}{x^2 + y^2 + z^2} dV \quad \dots\dots\dots (7 \text{ 分})$$

$$= \frac{1}{m} \int_0^{2\pi} d\theta \int_0^{\frac{\pi}{4}} d\varphi \int_0^{2\cos\varphi} \frac{r \cos\varphi}{r^2} r^2 \sin\varphi dr \quad \dots\dots\dots (9 \text{ 分})$$

$$= \frac{1}{m} 4\pi \int_0^{\frac{\pi}{4}} \sin\varphi \cos^3\varphi d\varphi = \frac{3}{4} \quad \dots\dots\dots (10 \text{ 分})$$

质心 $(0, 0, \frac{3}{4})$

九. $f(x) = (x^2 + 1) \sum_{n=0}^{\infty} \frac{(-1)^n}{2n+1} x^{2n+1}$ (3 分)

$$= \sum_{n=0}^{\infty} \frac{(-1)^n}{2n+1} x^{2n+3} + \sum_{n=0}^{\infty} \frac{(-1)^n}{2n+1} x^{2n+1} \dots\dots\dots (4 \text{ 分})$$

$$= \sum_{n=1}^{\infty} \frac{(-1)^{n-1}}{2n-1} x^{2n+1} + \sum_{n=0}^{\infty} \frac{(-1)^n}{2n+1} x^{2n+1} \dots\dots\dots (6 \text{ 分})$$

$$= x + \sum_{n=1}^{\infty} \left(\frac{(-1)^{n-1}}{2n-1} + \frac{(-1)^n}{2n+1} \right) x^{2n+1} = x + 2 \sum_{n=1}^{\infty} \frac{(-1)^{n-1}}{4n^2 - 1} x^{2n+1} \dots\dots\dots (8 \text{ 分})$$

收敛域为 $[-1, 1]$ (9 分)

十. 设曲面 $S_1: z = 1 \quad (x^2 + y^2 \leq 1)$

$$I = \oint_{S+S_1^-} - \iint_{S_1^-} (y^2 - x) dydz + (z^2 - y) dzdx + (x^2 - z) dxdy \dots\dots\dots (1 \text{ 分})$$

$$\oint_{S+S_1^-} (y^2 - x) dydz + (z^2 - y) dzdx + (x^2 - z) dxdy$$

$$= \iiint_V (-3) dV \dots\dots\dots (3 \text{ 分})$$

$$= -3 \int_0^{2\pi} d\theta \int_0^1 \rho d\rho \int_1^{2-\rho^2} dz \dots\dots\dots (4 \text{ 分})$$

$$= -\frac{3}{2} \pi \dots\dots\dots (5 \text{ 分})$$

$$\iint_{S_1^-} (y^2 - x) dydz + (z^2 - y) dzdx + (x^2 - z) dxdy$$

$$= \iint_{S_1^-} (x^2 - z) dxdy = - \iint_{D_{xy}} (x^2 - 1) dxdy \dots\dots\dots (6 \text{ 分})$$

$$= - \int_0^{2\pi} d\theta \int_0^1 \rho^3 \cos^2 \theta d\rho + \pi \dots\dots\dots (7 \text{ 分})$$

$$= \frac{3}{4} \pi \dots\dots\dots (8 \text{ 分})$$

$$I = -\frac{3}{2} \pi - \frac{3}{4} \pi = -\frac{9}{4} \pi \dots\dots\dots (9 \text{ 分})$$

十一. 当 $\lambda \neq 1$ 时, $\lim_{n \rightarrow \infty} b_n = 1 - \lim_{n \rightarrow \infty} \frac{\lambda \ln(1 + a_n)}{a_n} = 1 - \lambda \neq 0$

级数发散 (2 分)

当 $\lambda \neq 1$ 时, $\lim_{n \rightarrow \infty} \frac{b_n}{a_n} = \lim_{n \rightarrow \infty} \frac{a_n - \ln(1 + a_n)}{a_n^2}$ (3 分)

$= \lim_{n \rightarrow \infty} \frac{a_n - (a_n - \frac{1}{2}a_n^2 + o(a_n^2))}{a_n^2}$ (5 分)

$= \lim_{n \rightarrow \infty} \frac{\frac{1}{2}a_n^2 + o(a_n^2)}{a_n^2} = \frac{1}{2}$ (7 分)

因为 $\sum_{n=1}^{\infty} a_n$ 收敛, 故 $\sum_{n=1}^{\infty} b_n$ 也收敛 (8 分)